The Hellings-Downs curve for a Gaussian source

Viacheslav Chernosov

May 10, 2026

This paper is devoted to the derivation of the Hellings-Downs curve for the case of a Gaussian source of the Gravitational Wave Background.

1 Problem Statement

Consider the spherical coordinate system (ϕ, θ, r), where ϕ is longitude, θ is colatitude and r is radius. Then the orthonormal basis in the cartesian coordinate system will look like:

$$\hat{\Omega} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta), \quad (1a)$$
$$\hat{n} = (\cos\theta\cos\phi, \cos\theta\sin\phi, -\sin\theta), \quad (1b)$$
$$\hat{m} = (\sin\phi, -\cos\phi, 0), \quad (1c)$$

where $(\hat{\Omega}, \hat{n}, \hat{m}) \equiv (\hat{r}, \hat{\theta}, -\hat{\phi})$. Since $\bar{r} = r \cdot \hat{r}$, and the point on the unit sphere has the value $r = 1$, we can describe the unit sphere with a single vector $\hat{\Omega} = \hat{r}$.

We also know that gravitational waves have two polarization, which are called "plus" and "cross", and are described by the corresponding tensors:

$$e^{+}_{\alpha\beta}(\hat{\Omega}) = \hat{m}_\alpha \hat{m}_\beta - \hat{n}_\alpha \hat{n}_\beta, \quad (2a)$$
$$e^{\times}_{\alpha\beta}(\hat{\Omega}) = \hat{m}_\alpha \hat{n}_\beta + \hat{n}_\alpha \hat{m}_\beta \quad (2b)$$

Then, frequency shift of pulsar radiation $z = \Delta\nu/\nu_0$ in Fourier space:

$$\tilde{z}(f, \hat{\Omega}) = \left(e^{-i2\pi f L(1+\hat{\Omega}\cdot\hat{p})} - 1\right) h(f, \hat{\Omega}) F_{\hat{p}}(f, \hat{\Omega}), \quad (3)$$

where $\hat{p}$ is the unit vector in the direction of the pulsar, L is the distance to the pulsar, $h(f, \hat{\Omega})$ is spectrum of gravitational waves and $F_{\hat{p}}(\hat{\Omega})$ is pulsar response:

$$F_{\hat{p}}(\hat{\Omega}) = e_{\alpha\beta}(\hat{\Omega}) \frac{1}{2} \frac{\hat{p}^\alpha \hat{p}^\beta}{1 + \hat{\Omega}\cdot\hat{p}}. \quad (4)$$

Here we use a complex representation of the polarization tensor and spectrum, and in the case where the GWB is unpolarized, we have:

$$h(f, \hat{\Omega}) = h_+(f, \hat{\Omega}) + i \cdot h_\times(f, \hat{\Omega}) \quad (5)$$
$$e_{\alpha\beta}(\hat{\Omega}) = e^{+}_{\alpha\beta}(\hat{\Omega}) + i \cdot e^{\times}_{\alpha\beta}(\hat{\Omega}) \quad (6)$$

Since the Hellings-Downs curve relates correlations between signals from two pulsars, we defined the Hellings-Downs curve as:

$$\Gamma[\mathcal{P}](\hat{p}, \hat{q}) = \frac{3}{4\pi} \int_{S^2} d\hat{\Omega} \cdot \mathcal{K}(\hat{\Omega}, \hat{p}, \hat{q}) \cdot \mathcal{P}(\hat{\Omega}), \quad (7)$$

where $\mathcal{P}(\hat{\Omega})$ is the power spectrum on the unit sphere, and with the frequency power spectrum $\mathcal{H}(f)$ defined as:

$$|h(f, \hat{\Omega})|^2 = \mathcal{H}(f) \cdot \mathcal{P}(\hat{\Omega}), \quad (8)$$

and $\mathcal{K}(\hat{\Omega}, \hat{p}, \hat{q})$ is the integral kernel, which is defined as:

$$\mathcal{K}(\hat{\Omega}, \hat{p}, \hat{q}) = \mathcal{R}\left[F^*_{\hat{p}}(\hat{\Omega}) \cdot F_{\hat{q}}(\hat{\Omega})\right]. \quad (9)$$

And in our case, we assume that the power spectrum is Gaussian and looks like this:

$$\mathcal{P}(\hat{\Omega}) \sim \exp\left[-\frac{(\hat{\Omega} - \hat{\Omega}_0)^2}{2\sigma^2}\right], \quad (10)$$

where $\hat{\Omega}_0$ is the center of the source and σ is the width of the Gaussian. Converting the numerator:

$$(\hat{\Omega} - \hat{\Omega}_0)^2 = \hat{\Omega}^2 + \hat{\Omega}_0^2 - 2\hat{\Omega}\cdot\hat{\Omega}_0 = 1 + 1 - 2\hat{\Omega}\cdot\hat{\Omega}_0$$
$$(\hat{\Omega} - \hat{\Omega}_0)^2 = 2(1 - \hat{\Omega}\cdot\hat{\Omega}_0),$$

we have:

$$\mathcal{P}(\hat{\Omega}) \sim \exp\left[-\frac{2(1 - \hat{\Omega}\cdot\hat{\Omega}_0)}{2\sigma^2}\right],$$

$$\mathcal{P}(\hat{\Omega}) \sim \exp\left[-\frac{1}{\sigma^2}\right] \cdot \exp\left[\frac{\hat{\Omega}\cdot\hat{\Omega}_0}{\sigma^2}\right].$$

Let $\kappa = 1/\sigma^2$, and as a result we have:

$$\mathcal{P}(\hat{\Omega}) \sim \exp(-\kappa) \cdot \exp\left(\kappa\hat{\Omega}\cdot\hat{\Omega}_0\right). \quad (11)$$

And let's normalize our spectrum:

$$\frac{1}{4\pi} \cdot \int_{S^2} d\hat{\Omega} \cdot \mathcal{P}(\hat{\Omega}) = 1 \quad (12)$$

where:

$$\mathcal{P}(\hat{\Omega}) = G(\kappa) \cdot \exp\left(\kappa\hat{\Omega}\cdot\hat{\Omega}_0\right) \quad (13)$$

And to find the Hellings-Downs curve for a Gaussian source, we need to calculate:

$$\Gamma^{(g)}(\hat{p}, \hat{q}) = \frac{3 \cdot G(\kappa)}{4\pi} \int_{S^2} d\hat{\Omega} \cdot \mathcal{K}(\hat{\Omega}, \hat{p}, \hat{q}) \cdot e^{\kappa\hat{\Omega}\cdot\hat{\Omega}_0} \quad (14)$$

2 Integral in Coordinates

Firstly, let's represent the integral kernel $\mathcal{K}$ in vector form:

$$\mathcal{K} = \mathcal{R}\left[e^*_{\alpha\beta}(\hat{\Omega})\frac{1}{2}\frac{\hat{p}^\alpha\hat{p}^\beta}{1+\hat{\Omega}\cdot\hat{p}}\cdot e_{\mu\nu}(\hat{\Omega})\frac{1}{2}\frac{\hat{q}^\mu\hat{q}^\nu}{1+\hat{\Omega}\cdot\hat{q}}\right],$$

$$\mathcal{K} = \frac{1}{4}\frac{\hat{p}^\alpha\hat{p}^\beta\hat{q}^\mu\hat{q}^\nu}{(1+\hat{\Omega}\cdot\hat{p})(1+\hat{\Omega}\cdot\hat{q})}\mathcal{R}\left[e^*_{\alpha\beta}(\hat{\Omega})\cdot e_{\mu\nu}(\hat{\Omega})\right],$$

where:

$$\mathcal{R}\left[e^*_{\alpha\beta}(\hat{\Omega})\cdot e_{\mu\nu}(\hat{\Omega})\right] = e^+_{\alpha\beta}(\hat{\Omega})e^+_{\mu\nu}(\hat{\Omega}) + e^\times_{\alpha\beta}(\hat{\Omega})e^\times_{\mu\nu}(\hat{\Omega}),$$

then:

$$\mathcal{K} = \frac{1}{4}\frac{\hat{p}^\alpha\hat{p}^\beta\hat{q}^\mu\hat{q}^\nu\left[e^+_{\alpha\beta}(\hat{\Omega})e^+_{\mu\nu}(\hat{\Omega}) + e^\times_{\alpha\beta}(\hat{\Omega})e^\times_{\mu\nu}(\hat{\Omega})\right]}{(1+\hat{\Omega}\cdot\hat{p})(1+\hat{\Omega}\cdot\hat{q})}. \tag{15}$$

Consider individual tensor convolutions:

$$\hat{v}^\alpha\hat{v}^\beta e^+_{\alpha\beta}(\hat{\Omega}) = (\hat{v}\cdot\hat{m})^2 - (\hat{v}\cdot\hat{n})^2, \tag{16a}$$
$$\hat{v}^\alpha\hat{v}^\beta e^\times_{\alpha\beta}(\hat{\Omega}) = 2(\hat{v}\cdot\hat{m})(\hat{v}\cdot\hat{n}). \tag{16b}$$

Let's denote $P_a = (\hat{p}\cdot\hat{a})$ and $Q_a = (\hat{q}\cdot\hat{a})$, then:

$$\mathcal{K} = \frac{1}{4}\frac{(P_m^2 - P_n^2)\cdot(Q_m^2 - Q_n^2) + 4P_mP_nQ_mQ_n}{(1+P_\Omega)(1+Q_\Omega)}. \tag{17}$$

Note that since $(\hat{m},\hat{n},\hat{\Omega})$ is an orthonormal basis, and $|\hat{p}| = 1$, $|\hat{q}| = 1$, it means that:

$$P_n^2 = 1 - P_m^2 - P_\Omega^2,$$
$$Q_n^2 = 1 - Q_m^2 - Q_\Omega^2.$$

Then:

$$\frac{(P_n^2 - P_m^2)}{1+P_\Omega} = \frac{1 - 2P_m^2 - P_\Omega^2}{1+P_\Omega} = (1-P_\Omega) - \frac{2P_m^2}{1+P_\Omega},$$
$$\frac{(Q_n^2 - Q_m^2)}{1+Q_\Omega} = \frac{1 - 2Q_m^2 - Q_\Omega^2}{1+Q_\Omega} = (1-Q_\Omega) - \frac{2Q_m^2}{1+Q_\Omega}.$$

As a result, we can divide the integral kernel into four parts:

$$\mathcal{K}(\hat{\Omega},\hat{p},\hat{q}) = \frac{1}{4}(\mathcal{K}_I + \mathcal{K}_J + \mathcal{K}_K + \mathcal{K}_H), \tag{18}$$

where:

$$\mathcal{K}_I = (1-P_\Omega)(1-Q_\Omega), \tag{19a}$$
$$\mathcal{K}_J = -2\cdot\frac{P_m^2(1-Q_\Omega)}{(1+P_\Omega)}, \tag{19b}$$
$$\mathcal{K}_K = -2\cdot\frac{Q_m^2(1-P_\Omega)}{(1+Q_\Omega)}, \tag{19c}$$
$$\mathcal{K}_H = 4\cdot\frac{P_mQ_m(P_mQ_m + P_nQ_n)}{(1+P_\Omega)(1+Q_\Omega)}, \tag{19d}$$

and we can divide the integral into the same four parts:

$$\Gamma^{(g)}(\hat{p},\hat{q}) = \frac{3\cdot G(\kappa)}{4\pi}\cdot\frac{1}{4}\left(I + J + K + H\right), \tag{20}$$

where:

$$I = \int_{S^2} d\hat{\Omega}\cdot\mathcal{K}_I\cdot e^{\kappa\hat{\Omega}\cdot\hat{\Omega}_0} \tag{21a}$$
$$J = \int_{S^2} d\hat{\Omega}\cdot\mathcal{K}_J\cdot e^{\kappa\hat{\Omega}\cdot\hat{\Omega}_0} \tag{21b}$$
$$K = \int_{S^2} d\hat{\Omega}\cdot\mathcal{K}_K\cdot e^{\kappa\hat{\Omega}\cdot\hat{\Omega}_0} \tag{21c}$$
$$H = \int_{S^2} d\hat{\Omega}\cdot\mathcal{K}_H\cdot e^{\kappa\hat{\Omega}\cdot\hat{\Omega}_0} \tag{21d}$$

Let's define a convenient coordinate system. As we can see, all integrals include a term containing $(\hat{\Omega}\cdot\hat{\Omega}_0)$ in the exponent, and to simplify this term, we take $\hat{\Omega}_0 \equiv (0,0,1)$, because then $(\hat{\Omega}\cdot\hat{\Omega}_0) = \cos\theta$. For the pulsar vectors, we have:

$$\hat{p} = (\sin\alpha, 0, \cos\alpha) \tag{22a}$$
$$\hat{q} = (\sin\beta\cos\gamma, \sin\beta\sin\gamma, \cos\beta) \tag{22b}$$

where α, β are the angles between the source center and the pulsars, and γ is the angle between the directions to the pulsars from the source center. Now we can calculate all the scalar products:

$$P_\Omega = \sin\alpha\sin\theta\cos\phi + \cos\alpha\cos\theta,$$
$$Q_\Omega = \sin\beta\sin\theta\cos(\phi-\gamma) + \cos\beta\cos\theta,$$
$$P_n = \sin\alpha\cos\theta\cos\phi - \cos\alpha\sin\theta,$$
$$Q_n = \sin\beta\cos\theta\cos(\phi-\gamma) - \cos\beta\sin\theta$$
$$P_m = \sin\alpha\sin\phi,$$
$$Q_m = \sin\beta\sin(\phi-\gamma);$$

and define scalar product between $\hat{p}$ and $\hat{q}$ as:

$$\cos\xi = \sin\alpha\sin\beta\cos\gamma + \cos\alpha\cos\beta, \tag{23}$$

where ξ is the angle between $\hat{p}$ and $\hat{q}$.

Also, let's highlight the dependence of the coefficients on ϕ:

$$P_\Omega = a\cos\phi + b, Q_\Omega = c\cos(\phi-\gamma) + d, \tag{24a}$$
$$P_n = h\cos\phi - s, Q_n = k\cos(\phi-\gamma) - t, \tag{24b}$$
$$P_m = f\sin\phi, Q_m = g\sin(\phi-\gamma); \tag{24c}$$

where:

$$a = \sin\alpha\sin\theta, b = \cos\alpha\cos\theta, \tag{25a}$$
$$c = \sin\beta\sin\theta, d = \cos\beta\cos\theta, \tag{25b}$$
$$h = \sin\alpha\cos\theta, s = \cos\alpha\sin\theta, \tag{25c}$$
$$k = \sin\beta\cos\theta, t = \cos\beta\sin\theta, \tag{25d}$$
$$f = \sin\alpha, g = \sin\beta, \tag{25e}$$

3 Calculation of Integrals

All integrals are calculated first in longitude and then in latitude, and in all of them, after the first integration, we get an integral only in $\cos\theta$, and the integral can be replaced by an integral from -1 to $+1$ in x. Firstly, let's find the normalization $G(\kappa)$ of the spectrum:

$$\frac{1}{4\pi}\cdot\int_{S^2} d\hat{\Omega}\cdot G(\kappa)\cdot e^{\kappa\hat{\Omega}\cdot\hat{\Omega}_0}=1$$

$$\frac{1}{G(\kappa)}=\frac{1}{4\pi}\int_0^{\pi}\sin\theta\cdot d\theta\cdot e^{\kappa\cdot\cos\theta}\int_0^{2\pi}d\phi \tag{26}$$

$$\frac{1}{G(\kappa)}=\frac{1}{2}\cdot\int_0^{\pi}\sin\theta\cdot d\theta\cdot e^{\kappa\cdot\cos\theta} \tag{27}$$

Then we can rewrite the final integral as an integral in x:

$$\frac{1}{G(\kappa)}=\frac{1}{2}\int_0^{\pi}(-1)\cdot d(\cos\theta)\cdot e^{\kappa\cdot\cos\theta}=\frac{1}{2}\int_{-1}^{1}dx\cdot e^{\kappa x} \tag{28}$$

$$\frac{1}{G(\kappa)}=\frac{1}{2\kappa}\int_{-\kappa}^{\kappa}dy\cdot e^{y}=\frac{e^{\kappa}-e^{-\kappa}}{2\kappa}=\frac{\sinh(\kappa)}{\kappa}. \tag{29}$$

Next, let's calculate the integral of I using these steps:

$$I=\int_{S^2}d\hat{\Omega}\cdot\mathcal{K}_I\cdot e^{\kappa\hat{\Omega}\cdot\hat{\Omega}_0}$$

$$I=\int_0^{\pi}\sin\theta\cdot d\theta\cdot e^{\kappa\cdot\cos\theta}\int_0^{2\pi}d\phi\cdot\mathcal{K}_I \tag{30}$$

$$I_\phi=\int_0^{2\pi}d\phi\cdot\mathcal{K}_I \tag{31}$$

The result for I_ϕ is:

$$I_\phi=2\pi\cdot\left[(b-1)(d-1)+\frac{1}{2}\cdot ac\cos\gamma\right] \tag{32}$$

and after highlighting the dependency on $\cos\theta$:

$$\begin{aligned}I_\phi=2\pi\cdot\Big[&\cos^2\theta\cdot\left(\frac{3}{2}\cos\alpha\cos\beta-\frac{1}{2}\cos\xi\right)\\&-\cos\theta\cdot(\cos\alpha+\cos\beta)\\&+\left(1+\frac{1}{2}\cos\xi-\frac{1}{2}\cos\alpha\cos\beta\right)\Big]\end{aligned} \tag{33}$$

Then we can rewrite the final integral as an integral in x:

$$\begin{aligned}I&=\int_0^{\pi}(-1)\cdot d(\cos\theta)\cdot e^{\kappa\cdot\cos\theta}\cdot I_\phi(\cos\theta)\\&=\int_{-1}^{1}dx\cdot I_\phi(x)\cdot e^{\kappa x}.\end{aligned} \tag{34}$$

As a result, the integral I will be equal to:

$$\begin{aligned}I=4\pi\Big[&\frac{\sinh\kappa}{\kappa}\cdot(1+\cos\alpha\cos\beta)\\&+\frac{\kappa\cosh\kappa-\sinh\kappa}{\kappa^3}\cdot(\cos\xi-3\cos\alpha\cos\beta)\\&-\frac{\kappa\cosh\kappa-\sinh\kappa}{\kappa^2}\cdot(\cos\alpha+\cos\beta)\Big]\end{aligned} \tag{35}$$

Let's calculate the integral J using the same algorithm:

$$J=\int_{S^2}d\hat{\Omega}\cdot\mathcal{K}_J\cdot e^{\kappa\hat{\Omega}\cdot\hat{\Omega}_0}$$

$$J=\int_0^{\pi}\sin\theta\cdot d\theta\cdot e^{\kappa\cdot\cos\theta}\int_0^{2\pi}d\phi\cdot\mathcal{K}_J \tag{36}$$

$$J_\phi=\int_0^{2\pi}d\phi\cdot\mathcal{K}_J \tag{37}$$

The result for J_ϕ is:

$$\begin{aligned}J_\phi=4\pi f^2\cdot\Big[&\frac{(b+1)-\sqrt{(b+1)^2-a^2}}{a^2}\cdot(d-1)\\&+\left(\frac{(b+1)-\sqrt{(b+1)^2-a^2}}{a^2}\right)^2\cdot\frac{ac\cos\gamma}{2}\Big]\end{aligned} \tag{38}$$

and after highlighting the dependency on $\cos\theta$:

$$\begin{aligned}J_\phi=4\pi\cdot\Big[&\frac{1+\cos\alpha\cos\theta-|\cos\alpha+\cos\theta|}{1-\cos^2\theta}\cdot(\cos\beta\cos\theta-1)\\&+\frac{(1+\cos\alpha\cos\theta-|\cos\alpha+\cos\theta|)^2}{1-\cos^2\theta}\cdot\frac{\cos\xi-\cos\alpha\cos\beta}{2(1-\cos\alpha^2)}\Big]\end{aligned} \tag{39}$$

Then we can rewrite the final integral as an integral in x:

$$\begin{aligned}J&=\int_0^{\pi}(-1)\cdot d(\cos\theta)\cdot e^{\kappa\cdot\cos\theta}\cdot J_\phi(\cos\theta)\\&=\int_{-1}^{1}dx\cdot J_\phi(x)\cdot e^{\kappa x}\end{aligned} \tag{40}$$

As a result, the integral J will be equal to:

$$\begin{aligned}J=4\pi\Big[&\frac{(1-\cos\alpha)\cdot(1+\cos\alpha+\cos\beta+\cos\xi)}{1+\cos\alpha}\cdot\\&e^{-\kappa}\cdot(\mathrm{Ei}[\kappa(1-\cos\alpha)]-\mathrm{Ei}[2\kappa])\\&+\frac{(1+\cos\alpha)\cdot(1-\cos\alpha-\cos\beta+\cos\xi)}{1-\cos\alpha}\cdot\\&e^{\kappa}\cdot(\mathrm{Ei}[-\kappa(1+\cos\alpha)]-\mathrm{Ei}[-2\kappa])\\&+2\cdot\left(\cos\xi+\frac{\cos\alpha\cos\beta-\cos\xi}{1-\cos^2\alpha}\right)\cdot\\&\left(\frac{\cosh\kappa-\exp(-\kappa\cdot\cos\alpha)}{\kappa\cdot\cos\alpha}-\frac{\sinh\kappa}{\kappa}\right)\\&-(\cos\alpha\cos\beta-\cos\xi)\cdot\frac{\sinh\kappa}{\kappa}\Big]\end{aligned} \tag{41}$$

And since the integral K has the same structure, we can write the final equation for K by simply replacing $\cos\alpha$ and $\cos\beta$ in the equation for the integral J:

$$
\begin{aligned}
K = 4\pi\Big[&\frac{(1-\cos\beta)\cdot(1+\cos\alpha+\cos\beta+\cos\xi)}{1+\cos\beta}\cdot \\
&e^{-\kappa}\cdot(\mathrm{Ei}[\kappa(1-\cos\beta)]-\mathrm{Ei}[2\kappa]) \\
&+\frac{(1+\cos\beta)\cdot(1-\cos\alpha-\cos\beta+\cos\xi)}{1-\cos\beta}\cdot \\
&e^{\kappa}\cdot(\mathrm{Ei}[-\kappa(1+\cos\beta)]-\mathrm{Ei}[-2\kappa]) \\
&+2\cdot\left(\cos\xi+\frac{\cos\alpha\cos\beta-\cos\xi}{1-\cos^2\beta}\right)\cdot \\
&\left(\frac{\cosh\kappa-\exp(-\kappa\cdot\cos\beta)}{\kappa\cdot\cos\beta}-\frac{\sinh\kappa}{\kappa}\right) \\
&-(\cos\alpha\cos\beta-\cos\xi)\cdot\frac{\sinh\kappa}{\kappa}\Big]
\end{aligned} \tag{42}
$$

And finally, let's calculate the integral H:

$$
H = \int_{S^2} d\hat{\Omega}\cdot\mathcal{K}_H\cdot e^{\kappa\hat{\Omega}\cdot\hat{\Omega}_0}
$$

$$
H = \int_0^{\pi}\sin\theta\cdot d\theta\cdot e^{\kappa\cdot\cos\theta}\int_0^{2\pi}d\phi\cdot\mathcal{K}_H \tag{43}
$$

$$
H_\phi = \int_0^{2\pi}d\phi\cdot\mathcal{K}_H \tag{44}
$$

The result for H_ϕ is:

$$
H_\phi = 8\pi fg\cdot\left[A_H^{(1)}\cdot H_\phi^{(1)}+\frac{A_H^{(2)}\cdot H_\phi^{(2)}-A_H^{(3)}\cdot H_\phi^{(3)}}{B_H}\right] \tag{45}
$$

where:

$$
\begin{aligned}
A_H^{(1)} = \frac{ac\cos\gamma}{2}\cdot&\left[\frac{(b+1)-\sqrt{(b+1)^2-a^2}}{a^2}\right]\cdot \\
&\left[\frac{(d+1)-\sqrt{(d+1)^2-c^2}}{c^2}\right]
\end{aligned} \tag{46}
$$

$$
\begin{aligned}
A_H^{(2)} = ac\cos\gamma-&\left[(b+1)-\sqrt{(b+1)^2-a^2}\right]\cdot \\
&\left[(d+1)-\sqrt{(d+1)^2-c^2}\right]
\end{aligned} \tag{47}
$$

$$
A_H^{(3)} = ac\cos\gamma \tag{48}
$$

$$
\begin{aligned}
B_H = &\sqrt{(b+1)^2-a^2}\sqrt{(d+1)^2-c^2} \\
&-ac\cos\gamma+(b+1)(d+1)
\end{aligned} \tag{49}
$$

$$
\begin{aligned}
H_\phi^{(1)} = (hk+fg)\cdot\Big(&\frac{(b+1)-\sqrt{(b+1)^2-a^2}}{a^2} \\
&\cdot\frac{(d+1)-\sqrt{(d+1)^2-c^2}}{c^2}\Big) \\
&-4hk\frac{(b+1)(d+1)}{a^2c^2} \\
&-4fg\frac{\sqrt{(b+1)^2-a^2}\sqrt{(d+1)^2-c^2}}{a^2c^2}
\end{aligned} \tag{50}
$$

$$
\begin{aligned}
H_\phi^{(2)} = &th\frac{c(b+1)+a(d+1)\cos\gamma}{a^2c^2} \\
&+sk\frac{a(d+1)+c(b+1)\cos\gamma}{a^2c^2} \\
&+(hk-fg)\cdot\frac{(b+1)(d+1)}{a^2c^2}+st\frac{1}{ac}
\end{aligned} \tag{51}
$$

$$
\begin{aligned}
H_\phi^{(3)} = &th\frac{c\sqrt{(b+1)^2-a^2}+a\sqrt{(d+1)^2-c^2}\cos\gamma}{a^2c^2} \\
&+sk\frac{a\sqrt{(d+1)^2-c^2}+c\sqrt{(b+1)^2-a^2}\cos\gamma}{a^2c^2} \\
&-fg\cdot\frac{\sqrt{(b+1)^2-a^2}(d+1)+\sqrt{(d+1)^2-c^2}(b+1)}{a^2c^2}
\end{aligned} \tag{52}
$$

and after highlighting the dependency on $\cos\theta$:

$$
\begin{aligned}
H_\phi = 8\pi\cdot&(\cos\alpha\cos\beta-\cos\xi)\cdot \\
\Big[&\frac{1}{2}-\frac{1}{1-\cos\theta}\cdot\frac{(1-\cos\alpha\cos\beta)(\cos\alpha+\cos\beta)}{(1-\cos^2\alpha)(1-\cos^2\beta)} \\
&+\frac{C_H^{(1)}}{1-\cos^2\theta}\cdot\frac{1}{1-\cos^2\alpha}+\frac{C_H^{(2)}}{1-\cos^2\theta}\cdot\frac{1}{1-\cos^2\beta} \\
&+\frac{1-\cos\xi}{D_H}\cdot\left(1+\frac{C_H^{(1)}\cdot C_H^{(2)}}{(1-\cos^2\theta)(\cos\alpha\cos\beta-\cos\xi)}\right)\Big]
\end{aligned} \tag{53}
$$

where:

$$
C_H^{(1)} = |\cos\alpha+\cos\theta|-(\cos\alpha\cos\theta+1) \tag{54}
$$

$$
C_H^{(2)} = |\cos\beta+\cos\theta|-(\cos\beta\cos\theta+1) \tag{55}
$$

$$
\begin{aligned}
D_H = &|\cos\alpha+\cos\theta|\cdot|\cos\beta+\cos\theta| \\
&+(\cos\alpha\cos\theta+1)(\cos\beta\cos\theta+1) \\
&+(1-\cos^2\theta)(\cos\alpha\cos\beta-\cos\xi)
\end{aligned} \tag{56}
$$

Then we can rewrite the final integral as an integral in x:

$$
\begin{aligned}
H &= \int_0^{\pi}(-1)\cdot d(\cos\theta)\cdot e^{\kappa\cdot\cos\theta}\cdot H_\phi(\cos\theta) \\
&= \int_{-1}^{1}dx\cdot H_\phi(x)\cdot e^{\kappa x}
\end{aligned} \tag{57}
$$

As a result, the integral H will be equal to:

$$
\begin{aligned}
H = 4\pi\Big[& 2 \cdot (\cos\alpha\cos\beta - \cos\xi) \cdot \frac{\sinh\kappa}{\kappa} \\
& - \frac{(1-\cos\alpha)\cdot(1+\cos\alpha+\cos\beta+\cos\xi)}{1+\cos\alpha} \cdot \\
& e^{-\kappa} \cdot (\mathrm{Ei}[\kappa(1-\cos\alpha)] - \mathrm{Ei}[2\kappa]) \\
& - \frac{(1+\cos\alpha)\cdot(1-\cos\alpha-\cos\beta+\cos\xi)}{1-\cos\alpha} \cdot \\
& e^{\kappa} \cdot (\mathrm{Ei}[-\kappa(1+\cos\alpha)] - \mathrm{Ei}[-2\kappa]) \\
& - \frac{(1-\cos\beta)\cdot(1+\cos\alpha+\cos\beta+\cos\xi)}{1+\cos\beta} \cdot \\
& e^{-\kappa} \cdot (\mathrm{Ei}[\kappa(1-\cos\beta)] - \mathrm{Ei}[2\kappa]) \\
& - \frac{(1+\cos\beta)\cdot(1-\cos\alpha-\cos\beta+\cos\xi)}{1-\cos\beta} \cdot \\
& e^{\kappa} \cdot (\mathrm{Ei}[-\kappa(1+\cos\beta)] - \mathrm{Ei}[-2\kappa]) \\
& + (1-\cos\xi) \cdot \\
& 2 \cdot \mathcal{R}\left[e^{-\kappa\cdot r} \cdot (\mathrm{Ei}[\kappa(r-\cos\alpha)] + \mathrm{Ei}[\kappa(r-\cos\beta)])\right] \\
& - (1-\cos\xi) \cdot \\
& 2 \cdot \mathcal{R}\left[e^{-\kappa\cdot r} \cdot (\mathrm{Ei}[\kappa(r-1)] + \mathrm{Ei}[\kappa(r+1)])\right]\Big]
\end{aligned} \tag{58}
$$

where:

$$
r = \frac{\cos\alpha + \cos\beta + i\cdot\hat{S}}{1+\cos\xi}, \tag{59}
$$

and

$$
\begin{aligned}
\hat{S} &= \sqrt{1 + 2\cos\alpha\cos\beta\cos\xi - \cos^2\alpha - \cos^2\beta - \cos^2\xi} \\
&= |(\hat{p}\times\hat{q})\cdot\hat{\Omega}_0|
\end{aligned} \tag{60}
$$

Then the sum of the four integrals is:

$$
\begin{aligned}
I + J + K + H = 4\pi\Big[& \frac{\sinh\kappa}{\kappa} \cdot (1 + \cos\alpha\cos\beta) \\
& + \frac{\kappa\cosh\kappa - \sinh\kappa}{\kappa^3} \cdot (\cos\xi - 3\cos\alpha\cos\beta) \\
& - \frac{\kappa\cosh\kappa - \sinh\kappa}{\kappa^2} \cdot (\cos\alpha + \cos\beta) \\
& + 2 \cdot \left(\cos\xi + \frac{\cos\alpha\cos\beta - \cos\xi}{1-\cos^2\alpha}\right) \cdot \\
& \left(\frac{\cosh\kappa - \exp(-\kappa\cdot\cos\alpha)}{\kappa\cdot\cos\alpha} - \frac{\sinh\kappa}{\kappa}\right) \\
& + 2 \cdot \left(\cos\xi + \frac{\cos\alpha\cos\beta - \cos\xi}{1-\cos^2\beta}\right) \cdot \\
& \left(\frac{\cosh\kappa - \exp(-\kappa\cdot\cos\beta)}{\kappa\cdot\cos\beta} - \frac{\sinh\kappa}{\kappa}\right) \\
& + (1-\cos\xi) \cdot \\
& 2 \cdot \mathcal{R}\left[e^{-\kappa\cdot r} \cdot (\mathrm{Ei}[\kappa(r-\cos\alpha)] + \mathrm{Ei}[\kappa(r-\cos\beta)])\right] \\
& - (1-\cos\xi) \cdot \\
& 2 \cdot \mathcal{R}\left[e^{-\kappa\cdot r} \cdot (\mathrm{Ei}[\kappa(r-1)] + \mathrm{Ei}[\kappa(r+1)])\right]\Big]
\end{aligned} \tag{61}
$$

And the final integral $\Gamma^{(g)}(\hat{p},\hat{q})$ will be equal to:

$$
\Gamma^{(g)}(\hat{p},\hat{q}) = 1 + \frac{3(1-\cos\xi)}{2} \cdot \left[\mathcal{L}^{(g)}(\hat{p},\hat{q}) - \frac{1}{6}\right] + \frac{3}{2} \cdot \mathcal{D}^{(g)}(\hat{p},\hat{q}) \tag{62}
$$

where:

$$
\begin{aligned}
\mathcal{L}^{(g)}(\hat{p},\hat{q}) = \mathcal{R}\Big[& \frac{\kappa\cdot e^{-\kappa\cdot r}}{\sinh\kappa} \cdot \Big(\mathrm{Ei}[\kappa(r-\cos\alpha)] + \mathrm{Ei}[\kappa(r-\cos\beta)] \\
& - \mathrm{Ei}[\kappa(r-1)] - \mathrm{Ei}[\kappa(r+1)]\Big)\Big]
\end{aligned} \tag{63}
$$

$$
\begin{aligned}
\mathcal{D}^{(g)}(\hat{p},\hat{q}) = & \frac{\cos\beta - \cos\alpha\cos\xi}{1-\cos^2\alpha} \cdot \left[\coth\kappa - a\cdot\left(1 + \frac{e^{-\kappa a}}{a\sinh\kappa}\right)\right] \\
& + \frac{\cos\alpha - \cos\beta\cos\xi}{1-\cos^2\beta} \cdot \left[\coth\kappa - b\cdot\left(1 + \frac{e^{-\kappa b}}{b\sinh\kappa}\right)\right] \\
& + \frac{\cos\xi - 3\cos\alpha\cos\beta}{6} \cdot \left[\frac{3\coth\kappa}{\kappa} - \frac{3}{\kappa^2} - 1\right] \\
& - \frac{\cos\alpha + \cos\beta}{2} \cdot \left[\coth\kappa - \frac{1}{\kappa}\right]
\end{aligned} \tag{64}
$$

A How to calculate the integrals

For further calculations, we will use the standard integrals, where $k \in \mathbb{Z}$, $n, m \in \mathbb{N}_0$ and $-1 < a, c < 1$:

$$V_k(a) = \int_0^{2\pi} \frac{\cos k\phi \cdot d\phi}{a \cdot \cos\phi + 1} \tag{65}$$

$$W_m^n(a) = \int_0^{2\pi} \frac{\sin^n \phi \cdot \cos^m \phi \cdot d\phi}{a \cdot \cos\phi + 1} \tag{66}$$

$$Z(a, c, \gamma) = \int_0^{2\pi} \frac{d\phi}{(a \cdot \cos\phi + 1)(c \cdot \cos(\phi - \gamma) + 1)} \tag{67}$$

A.1 Integral V

We can find the integral $V_k(a)$ using a two-dimensional Poisson kernel:

$$P_r(\phi) = \sum_{n=-\infty}^{\infty} r^{|n|} e^{in\phi} = \frac{1 - r^2}{1 - 2r\cos\phi + r^2}, \tag{68}$$

where $0 \leq r < 1$. Let's rewrite the denominator in the same way:

$$1 + a \cdot \cos\phi = A(1 + r^2 - 2r\cos\phi) \tag{69}$$

$$\begin{cases} A \cdot (1 + r^2) = 1 \\ A \cdot (-2r) = a \end{cases}$$

$$A = \frac{1}{1 + r^2} \tag{70}$$

$$-\frac{2}{a} r = \frac{1}{A} = 1 + r^2$$

$$r^2 + \frac{2}{a} r + 1 = 0$$

$$r = \frac{-1 + \sqrt{1 - a^2}}{a} \tag{71}$$

Then we get:

$$\frac{1}{1 + a \cdot \cos\phi} = \frac{1 + r^2}{1 - r^2} \cdot \frac{1 - r^2}{1 + r^2 - 2r\cos\phi}$$

$$\frac{1}{1 + a \cdot \cos\phi} = \frac{1 + r^2}{1 - r^2} \sum_{n=-\infty}^{\infty} r^{|n|} e^{in\phi} \tag{72}$$

where:

$$\frac{1 + r^2}{1 - r^2} = -\frac{2r}{a} \cdot \frac{1}{-\frac{2r}{a} \cdot \sqrt{1 - a^2}} = \frac{1}{\sqrt{1 - a^2}}$$

And let's rewrite the numerator:

$$\cos k\phi = \frac{1}{2} \cdot (e^{ik\phi} + e^{-ik\phi})$$

Then the integral $V_k(a)$ will be equal to:

$$V_k(a) = \frac{1}{2\sqrt{1 - a^2}} \sum_{n=-\infty}^{\infty} r^{|n|} \int_0^{2\pi} e^{in\phi} \cdot (e^{ik\phi} + e^{-ik\phi}) \cdot d\phi$$

$$V_k(a) = \frac{1}{2\sqrt{1 - a^2}} \sum_{n=-\infty}^{\infty} r^{|n|} \int_0^{2\pi} (e^{i\phi(k+n)} + e^{i\phi(n-k)}) \cdot d\phi$$

$$V_k(a) = \frac{2\pi}{\sqrt{1 - a^2}} \sum_{n=-\infty}^{\infty} r^{|n|} \cdot \frac{\delta_{n,k} + \delta_{n,-k}}{2}$$

$$V_k(a) = \frac{2\pi}{\sqrt{1 - a^2}} \cdot \frac{r^{|k|} + r^{|-k|}}{2}$$

$$V_k(a) = \frac{2\pi}{\sqrt{1 - a^2}} \cdot \left(\frac{-1 + \sqrt{1 - a^2}}{a} \right)^{|k|} \tag{73}$$

A.2 Integral W

The first thing we notice is that for $n = 2k + 1$ the integral is zero, because:

$$W_m^{2k+1}(a) = \int_0^{2\pi} d\phi \cdot \sin\phi \cdot \frac{\sin^{2k}\phi \cdot \cos^m \phi}{a \cdot \cos\phi + 1}$$

$$= \int_0^{2\pi} d\phi \cdot \sin(-\phi) \cdot \frac{\sin^{2k}(-\phi) \cdot \cos^m(-\phi)}{a \cdot \cos(-\phi) + 1} = -W_m^{2k+1}(a)$$

$$W_m^{2k+1}(a) = 0 \tag{74}$$

and also, let's add a formula for reduction:

$$W_m^n(a) = W_m^{n-2}(a) - W_{m+2}^{n-2}(a), \tag{75}$$

because:

$$\sin^n(\phi)\cos^m(\phi) = \sin^{n-2}(\phi)\cos^m(\phi) - \sin^{n-2}(\phi)\cos^{m+2}(\phi) \tag{76}$$

Next, we can rewrite $\sin^{2k}\phi$ and $\cos^m\phi$ through $\cos k\phi$ like this:

$$\sin^{2k}(\phi) = \frac{\left(e^{i\phi} - e^{-i\phi}\right)^{2k}}{(2i)^{2k}} = \frac{(-1)^k}{2^{2k}} \cdot \sum_{s=0}^{2k} \binom{2k}{s} \cdot (e^{i\phi})^{2k-s}(-e^{-i\phi})^s$$

$$\sin^{2k}(\phi) = \frac{(-1)^k}{2^{2k}} \cdot \sum_{s=0}^{2k} \binom{2k}{s} \cdot (-1)^s \cdot e^{i\phi(2k-2s)} \tag{77}$$

$$\cos^m(\phi) = \frac{1}{2^m} \cdot \left(e^{i\phi} + e^{-i\phi}\right)^m = \frac{1}{2^m} \cdot \sum_{t=0}^{m} \binom{m}{t} (e^{i\phi})^{m-t}(e^{-i\phi})^t$$

$$\cos^m(\phi) = \frac{1}{2^m} \cdot \sum_{t=0}^{m} \binom{m}{t} \cdot e^{i\phi(m-2t)} \tag{78}$$

As a result, the numerator will look like:

$$\sin^{2k}(\phi)\cos^m(\phi) = \frac{(-1)^k}{2^{2k+m}} \cdot \sum_{s=0}^{2k}\sum_{t=0}^{m} \binom{2k}{s}\binom{m}{t} \cdot (-1)^s \cdot e^{i\phi(2k+m-2s-2t)} \tag{79}$$

And the integral will be equal to:

$$W_m^{2k}(a) = \frac{(-1)^k}{2^{2k+m}} \cdot \sum_{s=0}^{2k}\sum_{t=0}^{m} \binom{2k}{s}\binom{m}{t} \cdot (-1)^s \cdot V_{2k+m-2s-2t}(a) \tag{80}$$

where $V_{-k}(a) = V_k(a)$.

A.3 Integral Z

This integral is calculated in the complex plane, and the final result looks like this:

$$Z(a,c,\gamma) = \frac{2\pi}{1 - ac\cos\gamma + \sqrt{1-a^2}\sqrt{1-c^2}} \cdot \left(\frac{1}{\sqrt{1-a^2}} + \frac{1}{\sqrt{1-c^2}}\right) \tag{81}$$

Also note that any integral of the form:

$$Y^n_m(a,c,\gamma) = \int_0^{2\pi} \frac{\sin^n\phi \cdot \cos^m\phi \cdot d\phi}{(a\cdot\cos\phi + 1)(c\cdot\cos(\phi-\gamma)+1)} \tag{82}$$

Can be rewritten as:

$$Y^n_m(a,c,\gamma) = D\cdot Z(a,c,\gamma) + \sum_{k,s=0}^{N-2}\sum_{s=0}^{N-2} B^s_k \cdot W^s_k(0) + \sum_{k=0}^{N-1}\sum_{s=0}^{N-1} A^s_k \cdot W^s_k(a) + \sum_{k=0}^{N-1}\sum_{s=0}^{N-1} C^s_k \cdot W^s_k(c) \tag{83}$$

where $N = \max(n,m)$, because the numerator and denominator in Y^n_m are polynomials in $(\cos\phi,\ \sin\phi)$ and we can write:

$$\frac{P_N(x,y)}{L_1(x,y)\cdot L_2(x,y)} = \frac{P_0}{L_1(x,y)\cdot L_2(x,y)} + P_{N-2}(x,y) + \frac{P^{(1)}_{N-1}(x,y)}{L_1(x,y)} + \frac{P^{(2)}_{N-1}(x,y)}{L_2(x,y)} \tag{84}$$

where $x = \cos\phi$, $y = \sin\phi$, and:

$$\begin{cases} P_N(x,y) = x^n y^m \\ L_1(x,y) = xa + 1 \\ L_2(x,y) = xc\cos\gamma + yc\sin\gamma + 1 \end{cases} \tag{85}$$

And after reduction for $W^n_m(x)$, we can say that all our integrals are represented as:

$$K(a,c,\gamma) = D\cdot Z(a,c,\gamma) + \sum_{k}^{N-2} B'_k \cdot W^0_k(0) + \sum_{k=0}^{N-1} A'_k \cdot W^0_k(a) + \sum_{k=0}^{N-1} C'_k \cdot W^0_k(c) \tag{86}$$

Or in terms of $V_k(x)$:

$$K(a,c,\gamma) = D\cdot Z(a,c,\gamma) + B\cdot V_0(0) + \sum_{k=0}^{N-1}\left(A_k\cdot V_k(a) + C_k\cdot V_k(c)\right) \tag{87}$$

where $V_0(0) = 2\pi$. As the result, for the integrals I, J, K, H we will have:

$$I_\phi(a,b,c,d,\gamma) = 2\pi\cdot B^{(I)} \tag{88}$$

$$J_\phi(a,b,c,d,\gamma) = 2\pi\cdot\frac{B^{(J)}}{b+1} + \sum_{k=0}^{3}\frac{A^{(J)}_k}{b+1}\cdot V_k\left(\frac{a}{b+1}\right) \tag{89}$$

$$K_\phi(a,b,c,d,\gamma) = 2\pi\frac{B^{(K)}}{d+1} + \sum_{k=0}^{3}\frac{C^{(K)}_k}{d+1}\cdot V_k\left(\frac{c}{d+1}\right) \tag{90}$$

$$\begin{aligned} H_\phi(a,b,c,d,\gamma) = 2\pi\cdot&\frac{B^{(H)}}{(b+1)(d+1)} \\ +&\frac{D^{(H)}}{(b+1)(d+1)}\cdot Z\left(\frac{a}{b+1},\frac{c}{d+1},\gamma\right) \\ +&\sum_{k=0}^{3}\frac{A^{(H)}_k}{(b+1)(d+1)}\cdot V_k\left(\frac{a}{b+1}\right) \\ +&\sum_{k=0}^{3}\frac{C^{(H)}_k}{(b+1)(d+1)}\cdot V_k\left(\frac{c}{d+1}\right) \end{aligned} \tag{91}$$